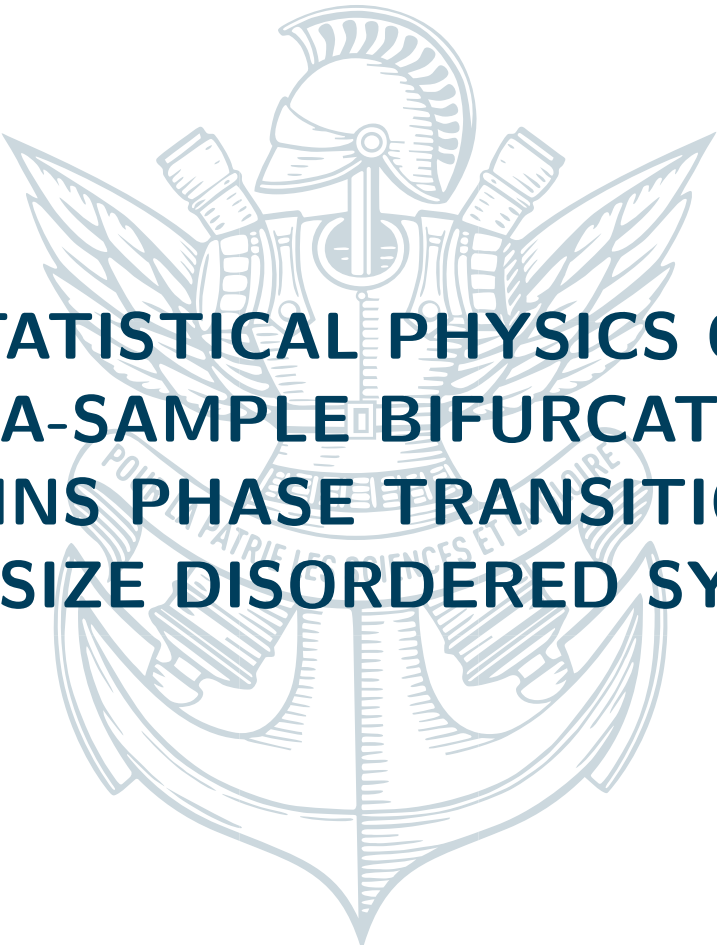




STATISTICAL PHYSICS OF INTRA-SAMPLE BIFURCATIONS EXPLAINS PHASE TRANSITIONS IN FINITE-SIZE DISORDERED SYSTEMS

Birth of Sequential Memory Recall in Delayed Non-Reciprocal Hopfield Networks

**Research Internship Report
James Franck Institute, University of Chicago**

March 23 – August 3, 2026

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Internship dates	March 23 – August 3, 2026
Confidentiality	Non-confidential

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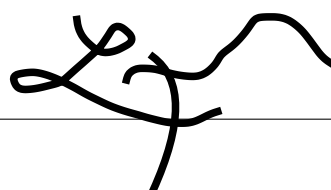
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ACKNOWLEDGMENTS

I would like to warmly thank Prof. Vincenzo Vitelli and Doruk Efe Gökmen for welcoming me to the James Franck Institute, helping me choose a productive research direction, and guiding me toward the right questions during the internship. I am also very grateful to Victor, a fellow intern from *École normale supérieure*, for sharing daily life in the laboratory and exploring this exciting new area of physics with me.

ABSTRACT

The Hopfield model describes associative memories as stable attractors of a recurrent neural network. Replacing part of its symmetric interaction by a non-reciprocal coupling from each memory to its successor makes ordered temporal recall possible, provided that the directed signal acts with a sufficient delay. During this internship, I studied the static-to-dynamic transition of this delayed mixed Hopfield network at extensive load, using sample-resolved numerical bifurcation analysis rather than a disorder-averaged description. I found that the transition of a finite network is built from an internal ensemble of memory-specific saddle-node bifurcations. The load controls the center of their near-Gaussian threshold distribution, while increasing system size narrows it in a central-limit-like manner. Two distinct collective boundaries select two different order statistics of this single distribution: its bulk marks the loss of extensive static recall, whereas its maximum predicts, realization by realization, the onset of the coherent sequential cycle. At the terminal barrier, the dynamics displays localized saddle-node slowing on a directly closed invariant recall route, consistent with a SNIC depinning mechanism. The released orbit recalls every stored binary pattern exactly and follows the additive relay clock $T_1 = \tau + t_{\text{esc}}$. I further characterized the intermediate hyperchaotic regime, showed how short delays or correlated patterns remove the ingredients required for localized depinning, and designed a multilayer Markovian architecture in which network depth generates an implicit delay. These results show how a single thermodynamic transition can conceal a finite-size statistical ensemble of local bifurcations that is directly observable within one disordered sample.

RÉSUMÉ

Le modèle de Hopfield décrit la mémoire associative comme un ensemble d'attracteurs stables d'un réseau de neurones récurrent. Remplacer une partie des interactions symétriques par un couplage non réciproque reliant chaque souvenir au suivant permet un rappel temporel ordonné, à condition que ce signal agisse avec un délai suffisant. Durant ce stage, j'ai étudié la transition entre rappel statique et rappel dynamique dans ce réseau de Hopfield mixte retardé à charge extensive, en résolvant numériquement les bifurcations de chaque réalisation du désordre. Dans un système fini, chaque mémoire disparaît à son propre seuil de bifurcation selle-nœud. Ces seuils forment, au sein d'un même échantillon, une distribution presque gaussienne dont la charge fixe le centre et dont la largeur décroît avec la taille selon un comportement proche d'un théorème central limite. Deux transitions collectives sélectionnent alors deux statistiques d'ordre différentes : le cœur de la distribution contrôle la fin du rappel statique extensif, tandis que son maximum prédit la naissance du cycle de rappel séquentiel. Au dernier obstacle, le ralentissement de type selle-nœud et la fermeture d'un cycle invariant sont cohérents avec un mécanisme de dépiégeage SNIC. Le cycle libéré rappelle exactement chaque motif binaire et suit l'horloge $T_1 = \tau + t_{\text{esc}}$. J'ai enfin caractérisé le régime hyperchaotique intermédiaire, testé le rôle du délai et des corrélations, et conçu une architecture multicouche dans laquelle la profondeur engendre un délai implicite.

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NOTATION AND CONVENTIONS

Symbol	Meaning
<i>Indices, memories, and neural dynamics</i>	
$i, j = 1, \dots, N$	Neuron indices; N is the number of neurons in each layer.
$\mu, \nu = 1, \dots, P$	Memory indices; P is the number of stored patterns.
$\alpha = P/N$	Memory load, held finite in the extensive-load limit.
ξ^μ, ξ_i^μ	Stored binary pattern and its components, with $\xi_i^\mu \in \{-1, +1\}$ and cyclic convention $\xi^{P+1} = \xi^1$.
$X \in \{-1, +1\}^{P \times N}$	Pattern matrix whose row μ is $(\xi^\mu)^\top$.
$\mathbf{u}(t), u_i(t)$	Continuous local-field vector and the field of neuron i .
$\phi(u), \phi(t)$	Scalar activation $\phi(u) = \tanh(\beta u)$ and its componentwise vector form $\phi(t) = \phi[\mathbf{u}(t)]$.
β	Neural gain; increasing β sharpens the activation toward the sign function.
t, t_0	Time and the single-neuron relaxation time used as the time unit.
J	Symmetric Hebbian coupling matrix that stabilizes static memories.
K	Non-reciprocal successor matrix that maps ξ^μ toward $\xi^{\mu+1}$.
S	Cyclic shift matrix acting on pattern indices, used in $K = N^{-1} X^\top S X$.
λ	Mixing fraction between symmetric reconstruction and delayed successor processing in the explicit-delay model.
τ	Explicit delay applied to the non-reciprocal interaction.
$\mathbf{a}(t), \mathbf{w}(t)$	Pattern-space coordinates and the transverse component in $\mathbf{u} = X^\top \mathbf{a} + \mathbf{w}$.
$m_\mu(t), b_\mu(t)$	Analog and binary overlaps with memory μ ; $\mathbf{m}(\mathbf{a})$ denotes the vector of analog overlaps in the reduced dynamics.
<i>Static branches and threshold statistics</i>	
$C(\lambda)$	Effective fixed-point coupling $(1 - \lambda)J + \lambda K$.
D	Diagonal local-gain matrix at a fixed point.
$z, \Delta(z)$	Characteristic growth rate and characteristic matrix of the delayed linearization.
I, I_P	Identity matrices in neural space and pattern space, respectively.
$\lambda_c(\mu)$	Terminal saddle-node threshold of the branch connected to memory μ .
$\bar{\lambda}_c, \sigma_{\lambda_c}$	Within-sample mean and standard deviation of the memory thresholds.
z_μ, Z_μ	Standardized threshold of memory μ and the corresponding standardized random variable in the position-scale description.
$\lambda_{50}(\mu)$	Coupling at which the successor component of branch μ reaches half its terminal value.
λ_*	Largest threshold in one sample, $\max_\mu \lambda_c(\mu)$.
$f_{\text{stat}}(\lambda)$	Fraction of the original retrieval branches that survive at λ .
Θ	Heaviside step function used to count surviving branches.
b_P	Finite- P Gaussian extreme-value correction for the expected maximum.

Continued on next page

Symbol	Meaning
<i>Sequential dynamics and stability</i>	
q	Slow coordinate in the saddle-node normal form.
$\lambda_{\text{onset}}^{\text{dyn}}$	Dynamically measured onset of the moving recall cycle.
λ_g	Ghost threshold inferred from the divergent relay-time fit above onset.
ζ_{max}	Leading real stability exponent of the pinned state.
t_{esc}	Nonlinear escape time through the remnant of a destroyed fixed point.
$T_1, \bar{T}$	Time between consecutive memory relays and mean period of a full tour.
D_{KY}	Finite-time Kaplan–Yorke dimension estimated from Lyapunov exponents.
d_{max}	Largest phase-aligned distance from the reference periodic trajectory.
<i>Implicit-delay multilayer model</i>	
k, K_{lay}	Layer index and total number of layers.
$\mathbf{U}_k(t)$	Neural field vector in layer k ; $\mathbf{U}_0$ is the first layer and $\mathbf{U}_{K_{\text{lay}}-1}$ is the readout layer used to measure memory transitions and recall periods.
λ_{ff}	Feed-forward coupling between successive layers, distinct from the mixing fraction λ of the explicit-delay model.
$a(\lambda_{\text{ff}}), b(\lambda_{\text{ff}})$	Intercept and per-layer slope of the implicit-delay recall-period law.
<i>Correlated-pattern test</i>	
c, s_i^μ	Target nearest-neighbor correlation and the binary flip variable used to construct a correlated Markov sequence.
q_1	Measured nearest-neighbor overlap in the smooth periodic construction.
K_{F}	Number of retained Fourier modes in the smooth periodic construction; this quantity is printed as K inside Figure 12 and is unrelated to the successor matrix K .

Bold symbols denote vectors. Unless stated otherwise, numerical values refer to a reference realization with $N = 2000$, $P = 100$, $\alpha = 0.05$, $\beta = 20$, $\tau = 10t_0$, and disorder seed 42. The coupling λ_{ff} is used only for the multilayer model in [Section 7](#).

1

INTRODUCTION: FROM MEMORIES TO MOTION

A stored memory is useful only if it can be retrieved. In a conventional computer, this operation is performed through an address: one specifies where information was stored and reads it back. Biological memory is different. A partial smell, image, or sound can reactivate a complete experience, even when no explicit address is available. The Hopfield network gave this idea a physical formulation: a memory can be a stable state of a many-body dynamical system, and recall can be the relaxation toward that state from an incomplete initial condition [1].

This static picture immediately raises a second question. Many computations are not single configurations but ordered processes: a motor program, a melody, or a sequence of steps in an algorithm. How can a recurrent network store individual states and also move autonomously through them in a prescribed order? Classical work showed that a symmetric Hebbian interaction can stabilize each memory, while an asymmetric interaction from one memory to the next can encode temporal association. A sufficiently slow synaptic response or an explicit delay then gives the network time to express one state before advancing to its successor [2, 3]. What remains less understood is the dynamical mechanism by which static recall gives way to coherent motion.

That mechanism was the subject of my research internship at the James Franck Institute at the University of Chicago, in Vincenzo Vitelli's group and under the supervision of Vincenzo Vitelli and Doruk Efe Gökmen. The group studies collective phenomena far from equilibrium, with particular emphasis on non-reciprocal interactions: interactions for which the influence of degree of freedom i on j is not the reverse of the influence of j on i [4]. Such systems do not generally relax in an energy landscape and can support phases whose order is intrinsically dynamical.

Efe proposed that I study a mixed Hopfield network motivated by a two-population model on which he was working. The effective one-population description combines a symmetric storage term with a non-reciprocal successor term. Following the established physics of temporal association, the successor signal is delayed. I started by studying the statistical mechanics of static and dynamical Hopfield networks and by reproducing classical results for discrete models. I also constructed preliminary phase diagrams using discrete-time simulations. My initial goal was to derive a dynamical mean-field theory (DMFT) for the mixed network. Rather than spend the internship on a long formal derivation, however, I chose to move to the continuous-time system and ask a question that could be answered directly: *which bifurcations build the transition from static memory to sequential recall?*

A conventional phase diagram assigns one boundary to a realization or to its disorder average. By continuing the equilibrium connected to each memory separately, I found instead that a finite network contains an internal ensemble of critical points. Every memory branch terminates at its own quenched saddle-node threshold $\lambda_c(\mu)$. The central part of this distribution controls the loss of an extensive number of static memories, whereas its maximum

$$\lambda_* = \max_{1 \leq \mu \leq P} \lambda_c(\mu) \quad (1)$$

controls the release of the global sequence-retrieving orbit. One microscopic distribution therefore produces two macroscopic boundaries at finite size.

The main contributions of this internship are:

1. an exact sample-preserving reduction of the neural dynamics onto the space generated by the stored patterns;
2. the numerical continuation and stability analysis of all memory-connected equilibria, revealing a near-Gaussian ensemble of memory-specific folds;
3. the identification, across independent disorder realizations, of the largest static threshold with the onset and bottleneck of sequential recall;
4. the characterization of localized saddle-node slowing on a directly closed terminal invariant recall route, consistent with a SNIC depinning mechanism;

5. the discovery of exact binary recall and of the additive relay clock $T_1 = \tau + t_{\text{esc}}$;
6. tests showing that the mechanism requires both a localized propagating front and discrete heterogeneous barriers;
7. the design of a multilayer Markovian architecture in which depth replaces the explicit delay by an implicit dynamical memory.

The report is organized to make these results accessible without prior knowledge of neural-network statistical mechanics. [Section 2](#) introduces classical Hopfield networks, temporal association, DMFT, and the language of bifurcations. [Section 3](#) defines the model and the numerical strategy. The main results are then developed from the static branches ([Section 4](#)) to the global cycle ([Section 5](#)), before testing the ingredients of the mechanism ([Section 6](#)) and replacing the explicit delay by network depth ([Section 7](#)). The final sections assess the limits of the present work and the route toward an analytical theory.

Take-home message. The classical model explains how delayed non-reciprocity can generate a sequence. The new question addressed here is how a finite disordered realization makes the transition from many static attractors to one coherent moving orbit.

2

SCIENTIFIC BACKGROUND AND STATE OF THE ART

This section builds the minimum background needed for the rest of the report. The point of view comes from physics: we first turn memories into attractors, then ask what symmetric interactions can and cannot do, and finally introduce the non-equilibrium tools required for a moving state.

2.1 FROM A NEURON TO A COLLECTIVE STATE

A biological neuron integrates many inputs and emits an electrical response only when their combined influence is sufficiently strong. Neural-network models retain this input-output logic while discarding most cellular detail [5]. A neuron is represented by a state s_i or a continuous local field u_i . The function $\phi(u_i)$ converts that field into an effective firing activity. In the present report,

$$\phi(u) = \tanh(\beta u), \tag{2}$$

where the gain β controls how sharply the response switches between negative and positive activity. For large β , $\tanh(\beta u)$ approaches the binary sign function.

The essential feature is recurrence. If neuron j influences neuron i through a coupling J_{ij} , then the activity of i changes the future of the same network that produced it. A state is a fixed point when this feedback reproduces the state itself. A memory can therefore be encoded by choosing the couplings so that the desired neural configuration becomes a stable fixed point.

2.2 THE BINARY HOPFIELD MODEL

Consider N binary neurons $s_i \in \{-1, +1\}$ and P random binary patterns ξ^μ , $\mu = 1, \dots, P$. The standard Hebbian prescription is

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu, \quad J_{ij} = J_{ji}. \quad (3)$$

Two neurons that have the same sign in many stored patterns receive a positive coupling; two neurons that often disagree receive a negative one. Under asynchronous zero-noise dynamics, one neuron at a time adopts the sign of its local field,

$$s_i \leftarrow \text{sgn} \left(\sum_j J_{ij} s_j \right). \quad (4)$$

Because the couplings are symmetric, this dynamics decreases the energy

$$E(\mathbf{s}) = -\frac{1}{2} \sum_{ij} J_{ij} s_i s_j. \quad (5)$$

A single spin update changes E by an amount whose sign is fixed by the alignment of the spin with its local field. Unless the spin is already aligned, the energy decreases. The dynamics must therefore end in a local minimum: recurrent feedback has been turned into descent on an energy landscape [1, 6].

The quality of recall is quantified by the overlap

$$m_\mu = \frac{1}{N} \sum_{i=1}^N \xi_i^\mu s_i. \quad (6)$$

If $m_\mu = 1$, the network exactly matches pattern μ ; if $m_\mu \simeq 0$, it is essentially uncorrelated with it. An initial condition with a positive but imperfect overlap can flow toward the stored pattern. This is content-addressable memory: the input describes part of the content rather than the location at which it was stored.

The resulting attractor picture is summarized schematically in Figure 1.

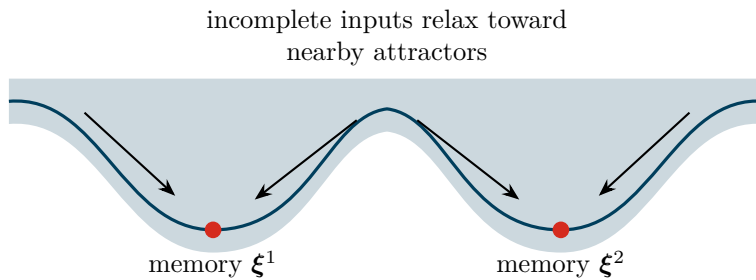


Figure 1: Associative memory as motion in an energy landscape. Symmetric couplings make the dynamics descend toward local minima. Stored patterns are designed to be broad minima, so corrupted inputs within their basins are corrected during the relaxation. The picture is schematic: the true landscape lives in an N -dimensional configuration space.

2.3 CROSS-TALK, SPURIOUS STATES, AND THE EXTENSIVE-LOAD TRANSITION

The Hebbian construction is not exact when many patterns are stored. Evaluating the local field near pattern μ separates a desired term proportional to ξ_i^μ from a sum of contributions due to all other patterns. This second part is the *cross-talk noise*. When P remains finite as $N \rightarrow \infty$, random pattern overlaps vanish and recall is almost perfect. The physically richer regime is

$$P = \alpha N, \quad (7)$$

where the number of stored memories grows with the network. Infinitely many individually small overlaps then combine into a finite random field.

Amit, Gutfreund, and Sompolinsky used the statistical mechanics of disordered magnetic systems to analyze this regime [7–9]. At low load, the energy landscape contains retrieval states with one macroscopic overlap. As the load or noise increases, mixtures and spin-glass-like states proliferate. For random unbiased patterns at zero temperature, the familiar thermodynamic threshold for typical retrieval is approximately $\alpha_c \simeq 0.138$.

This classical transition is already a warning for the present work. In a finite system, the random cross-talk is not identical for every pattern. Even if memories are equivalent after disorder averaging, their realized local environments need not be. My analysis will show that this inequivalence splits one averaged transition into many sample-specific bifurcations.

2.4 CONTINUOUS-TIME HOPFIELD DYNAMICS

Hopfield extended the binary model to graded neurons obeying a relaxation equation [10]:

$$t_0 \dot{u}_i(t) = -u_i(t) + \sum_j J_{ij} \phi[u_j(t)]. \quad (8)$$

The first term makes each field forget its past on the time scale t_0 ; the recurrent input reconstructs the collective state. For monotone ϕ and symmetric interactions, one can again construct a Lyapunov function. The detailed trajectory differs from the binary asynchronous model, but the long-time computation remains static: stable attractors are fixed points.

Continuous dynamics is essential for this work because it separates three time scales (neuronal relaxation, propagation delay, and nonlinear escape from a memory) and makes the transition accessible to the tools of bifurcation theory.

2.5 TEMPORAL ASSOCIATION: ASYMMETRIC COUPLINGS AND SLOW RESPONSE

To encode a prescribed sequence [11], a natural successor matrix is

$$K_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^{\mu+1} \xi_j^\mu. \quad (9)$$

Activity aligned with ξ^μ is then mapped toward $\xi^{\mu+1}$. Unlike J , the matrix K is not symmetric. Detailed balance and the Hopfield energy description are lost, but an orientation appears in state space.

The combination of symmetric storage, asymmetric successor couplings, and slow synaptic response is classical. Sompolinsky and Kanter showed that it can stabilize temporal associations and cycles of patterns, with the slow response setting the residence time in each state [2, 12]. Kleinfeld independently developed a closely related mechanism for sequential state generation [3]; later work connected it to biological central pattern generators [13] and temporally asymmetric Hebbian rules [14]. Mixed symmetric-asymmetric networks and the competition between reconstruction and sequence processing were also studied in discrete dynamics [15–19].

2.6 DYNAMICAL MEAN-FIELD THEORY: THE INITIAL ANALYTICAL ROUTE

Equilibrium statistical mechanics relies on an energy or Gibbs distribution, neither of which is generally available for asymmetric dynamics. Dynamical mean-field theory attacks the trajectories directly. The generating-functional construction introduces sources for the neural histories, averages over the quenched disorder, and takes the large- N saddle point. The many-neuron problem becomes an effective single-site stochastic process whose noise covariance and response kernel must agree self-consistently with the correlation and response of the process itself [20, 21].

During the first part of the internship, I studied this formalism and reproduced derivations for classical models. A full DMFT of the delayed mixed system would be a natural analytical continuation of the project. It was not required to discover the finite-size mechanism, however: disorder averaging would in fact erase the intra-sample threshold distribution that became the central object of the work.

2.7 BIFURCATIONS AS A MICROSCOPE FOR PHASE TRANSITIONS

A bifurcation is a qualitative change of a dynamical state when a control parameter is varied [22, 23]. Three elementary scenarios will recur:

- At a *saddle-node* or fold, a stable fixed point collides with an unstable one and disappears. A real stability eigenvalue reaches zero.
- At a *Hopf bifurcation*, a complex-conjugate pair crosses the imaginary axis and a finite-frequency oscillation is created or destroyed.
- In a *saddle-node on an invariant circle* (SNIC), the colliding fixed points lie on a closed invariant route. Their disappearance releases a periodic orbit whose period diverges at onset.

Near a generic saddle-node, a slow coordinate q follows the normal form

$$\dot{q} = A(\lambda - \lambda_c) + Bq^2. \quad (10)$$

Above threshold, integrating through the resulting bottleneck gives an escape time $t_{\text{esc}} \propto (\lambda - \lambda_c)^{-1/2}$. Closely related bottleneck laws occur when fronts depin from discrete obstacles [24, 25]. This square-root slowing identifies the local collision. Calling it a SNIC additionally requires evidence that the collision belongs to a closed global route.

Delay equations have the extra subtlety that their state is an entire history segment, not only the instantaneous vector [26, 27]. The basic physical diagnostics nevertheless remain intuitive: continuation locates fixed points, the characteristic spectrum tests their stability, Floquet multipliers test a periodic orbit, and Lyapunov exponents distinguish regular from chaotic motion.

The boundary between these established ingredients and the results obtained during this internship is summarized in Table 1.

Table 1: Boundary between prior work and the contribution of this internship. The model ingredients have classical precedents; the contribution concerns their sample-resolved finite-size bifurcation structure and its dynamical consequences.

Established before this internship	Contribution of this work
Symmetric Hebbian couplings store patterns as fixed-point attractors.	Continuation of the equilibrium connected to every stored pattern in one realized network.
Asymmetric successor couplings encode a prescribed temporal order.	Identification of a memory-specific saddle-node threshold for every branch.
Slow or delayed interactions allow sequential recall.	Separation of the loss of extensive static recall from the onset of the coherent cycle.
Mean-field theories describe disorder-averaged phase diagrams.	A quenched intra-sample threshold distribution whose bulk and maximum control different observables.
Time delay is needed for sequential recall in continuous-time dynamics.	Identification of the extreme memory as the localized bottleneck and direct closure of its terminal recall route.
	A concrete multilayer architecture whose depth generates the sequential-recall clock without an explicit delay.

Take-home message. The delayed mixed Hopfield architecture is classical. The contribution of this internship is to resolve, within individual finite samples, the ensemble of local bifurcations that constructs the global static-to-dynamic transition.

3

RESEARCH QUESTION, MODEL, AND APPROACH

My objective was to resolve the transition between two competing functions: Hebbian static recall and non-reciprocal sequence retrieval. At low non-reciprocity, the network should remain near a static memory; at high non-reciprocity, it should travel around the stored sequence. The central question is what happens between these limits in one finite disordered realization.

3.1 DELAYED MIXED HOPFIELD DYNAMICS

I consider N continuous neurons with local fields $u_i(t)$ and firing rates $\phi_i(t) = \tanh[\beta u_i(t)]$. Their evolution is

$$t_0 \dot{\mathbf{u}}(t) = -\mathbf{u}(t) + (1 - \lambda)J\boldsymbol{\phi}(t) + \lambda K\boldsymbol{\phi}(t - \tau). \quad (11)$$

Here t_0 is the single-neuron relaxation time, τ is a transmission delay, and $\lambda \in [0, 1]$ mixes two interactions. For unbiased random patterns $\boldsymbol{\xi}^\mu \in \{-1, +1\}^N$, with $\boldsymbol{\xi}^{P+1} = \boldsymbol{\xi}^1$,

$$J = \frac{1}{N} \sum_{\mu=1}^P \boldsymbol{\xi}^\mu (\boldsymbol{\xi}^\mu)^\top, \quad K = \frac{1}{N} \sum_{\mu=1}^P \boldsymbol{\xi}^{\mu+1} (\boldsymbol{\xi}^\mu)^\top. \quad (12)$$

The two terms perform complementary operations. The symmetric matrix J reconstructs the currently expressed memory. The non-reciprocal matrix K maps that memory toward its successor. The delayed state is

seen only by K , so the delay changes the temporal drive without changing which recurrent patterns are encoded by J . The parameter λ therefore interpolates between standing memory and directed motion.

The architecture and its two dynamical regimes are illustrated in Figure 2.

Two overlaps provide complementary readouts. The graded overlap

$$m_\mu(t) = \frac{1}{N}(\xi^\mu)^\top \tanh[\beta \mathbf{u}(t)] \quad (13)$$

measures analog alignment, while

$$b_\mu(t) = \frac{1}{N}(\xi^\mu)^\top \text{sgn}[\mathbf{u}(t)] \quad (14)$$

asks whether the signs of all neurons reproduce the binary memory. Unless specified otherwise, branch-resolved results refer to

$$N = 2000, \quad P = 100, \quad \alpha = 0.05, \quad \beta = 20, \quad \tau = 10t_0, \quad (15)$$

with disorder seed 42.

A Hopfield network that can also replay its memories in order

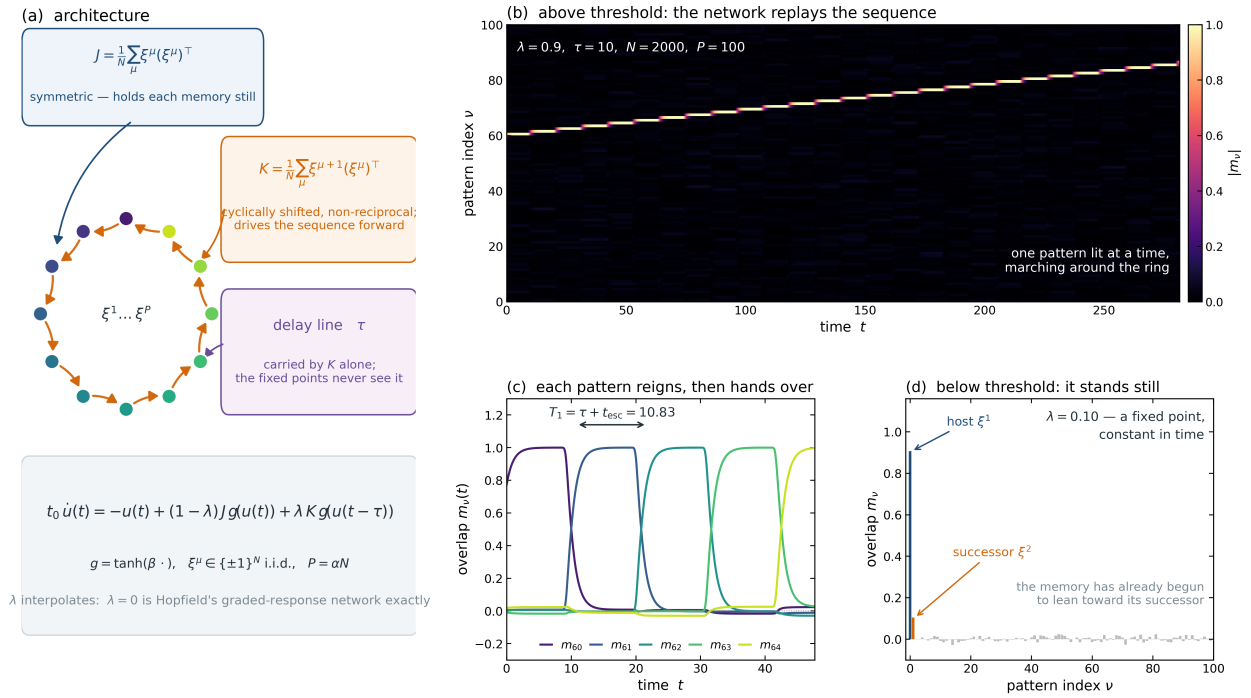


Figure 2: A Hopfield network that can also replay its memories in order. (a) Symmetric Hebbian couplings stabilize the stored patterns; a delayed non-reciprocal coupling maps each pattern to its cyclic successor. (b) Above the collective onset, one localized activity peak travels around the memory ring. (c) Consecutive overlaps show a relay clock: each memory is expressed and then hands activity to the next. (d) Below onset, the same directed coupling only tilts a fixed point toward its successor. The small gray bars in panel (d) indicate the typical cross-talk scale schematically and are not separate measurements of each overlap.

3.2 AN EXACT REDUCTION THAT PRESERVES THE SAMPLE

Directly evolving all N neurons is unnecessary because the interactions have rank at most P . Let $X \in \{-1, +1\}^{P \times N}$ contain the patterns as rows, and let S shift a pattern index to its cyclic successor. Then

$$J = \frac{1}{N} X^\top X, \quad K = \frac{1}{N} X^\top S X. \quad (16)$$

For $P < N$, a random pattern matrix has full row rank with overwhelming probability; this condition was also verified for every realization used here. It makes the pattern-space coordinates unique. Decompose the neural state as

$$\mathbf{u} = X^\top \mathbf{a} + \mathbf{w}, \quad \mathbf{w} \perp \text{range}(X^\top). \quad (17)$$

Both recurrent inputs in Equation (11) lie in the pattern subspace. The transverse component therefore obeys exactly

$$t_0 \dot{\mathbf{w}} = -\mathbf{w}. \quad (18)$$

It decays exponentially, so the pattern subspace is invariant and globally attracting. Within it, the dynamics closes on P delayed coordinates:

$$t_0 \dot{\mathbf{a}}(t) = -\mathbf{a}(t) + (1 - \lambda) \mathbf{m}[\mathbf{a}(t)] + \lambda S \mathbf{m}[\mathbf{a}(t - \tau)], \quad (19)$$

where

$$\mathbf{m}(\mathbf{a}) = \frac{1}{N} X \tanh(\beta X^\top \mathbf{a}). \quad (20)$$

This is not a mean-field approximation. The nonlinear map is computed with the realized pattern matrix X , so every finite-size cross-overlap is retained. The reduction removes directions that decay trivially while preserving the quenched fluctuations that split the memory thresholds. Full and reduced trajectories agree to within 5.5×10^{-16} on the certification trajectory. The result turns a network with N neural variables into P scalar delay equations, although the dynamical state still includes their history over the interval $[t - \tau, t]$.

3.3 NUMERICAL STRATEGY

The analysis combines several measurements that answer logically independent questions. First, I continued the equilibrium connected to each pattern from $\lambda = 0$ until the branch ended. A direct stability calculation distinguished a real saddle-node from an oscillatory Hopf instability. Second, I integrated the delayed dynamics from static and moving initial histories to locate attractors, measure memory relays, and compare the static thresholds with the onset of global motion. Third, I used Floquet multipliers to test the stability of periodic recall and Lyapunov exponents to characterize irregular trajectories. Finally, I repeated the central measurements over disorder realizations and system sizes rather than relying on one visually compelling trajectory.

The low-rank structure of Equation (16) is used throughout: neither the dense $N \times N$ couplings nor their dense Jacobians need be stored. Fixed points and their folds were computed in double precision, and critical thresholds were bracketed by bisection. Time integration and continuation tolerances were refined until the claimed effects were much larger than numerical uncertainty.

3.4 PHENOMENOLOGICAL PHASE ORGANIZATION

Before resolving individual bifurcations, I scanned the (α, λ) plane at three delays. Starting from a stored pattern, trajectories were classified as persistent static recall, ordered forward relays, or failure of both criteria. The result in Figure 3 shows the broad organization. Longer delays open a wide region of coherent sequence processing; at short delay, irregular dynamics dominates.

The smooth boundary visible at this scale hides the finite-size structure. As λ increases, different static memories disappear at different values; the coherent cycle is not released until the last relevant obstruction has vanished. The next section resolves that hidden sequence of events.

Take-home message. The model is a competition between symmetric reconstruction and delayed directed motion. Its low-rank form makes every realized memory branch accessible without averaging away the finite-size disorder.

4

A QUENCHED ENSEMBLE OF MEMORY BIFURCATIONS

The phase diagram of Figure 3 suggests one boundary between static and moving recall. Looking at individual memories changes that picture. In one finite network, the P retrieval states are statistically equivalent before the patterns are drawn, but no longer equivalent after the disorder has been realized. Each memory sees its own cross-talk environment and disappears at its own value of the non-reciprocal fraction λ .

This section develops that observation in three steps. I first establish that the memory branches terminate through saddle-node bifurcations rather than delay-induced oscillatory instabilities. I then resolve the entire set of thresholds in one sample. Finally, I show that these thresholds form a simple statistical ensemble whose typical and extreme values anticipate two different collective transitions.

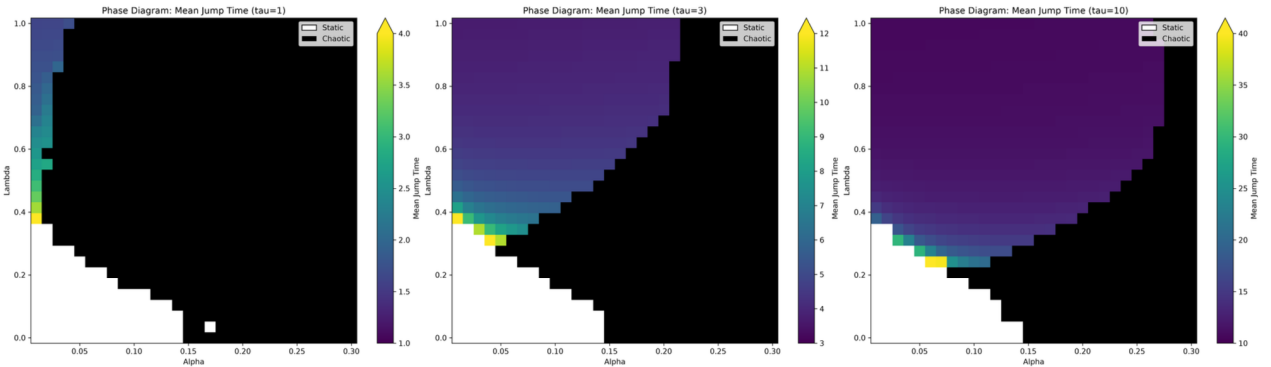


Figure 3: Phenomenological static-to-sequential phase organization. Operational phase diagrams in the (α, λ) plane for $\tau/t_0 = 1, 3, 10$ (left to right). White points retain the initial memory; colored points advance through the prescribed sequence and are shaded by their mean relay time; black points meet neither criterion. The survey used the continuous-time model, $N = 2500$, $\beta = 20$, a 30×30 grid, Euler integration with $\Delta t = 0.05t_0$, and zero diagonal couplings. It motivates the detailed analysis but is not used to estimate the thresholds below.

The full picture, from individual folds to the released cycle and the narrowing of the threshold distribution, is summarized in Figure 4.

4.1 WHY THE TERMINAL EVENTS ARE STATIC FOLDS

At a fixed point, the present and delayed activities are identical. Equation (11) therefore reduces to

$$\mathbf{F}(\mathbf{u}; \lambda) = -\mathbf{u} + C(\lambda) \tanh(\beta \mathbf{u}) = \mathbf{0}, \quad C(\lambda) = (1 - \lambda)J + \lambda K. \quad (21)$$

The delay τ does not appear. Consequently, the geometry of every equilibrium branch and the location of every fold are exactly delay-independent. The delay may still alter the stability of a branch before its fold, so a possible Hopf bifurcation must be ruled out separately.

Let

$$D = \beta \operatorname{diag}\{\operatorname{sech}^2(\beta u_i^*)\} \quad (22)$$

be the local gain matrix at a fixed point. A perturbation proportional to $e^{zt}\mathbf{v}$ is governed by the characteristic matrix

$$\Delta(z) = (t_0 z + 1)I - (1 - \lambda)JD - \lambda e^{-z\tau}KD. \quad (23)$$

The exponential factor is the mathematical trace of the delay: unlike an ordinary Jacobian, this equation has infinitely many roots. The low rank of the couplings again makes the calculation tractable. Using Sylvester's

From memory-specific folds to collective recall

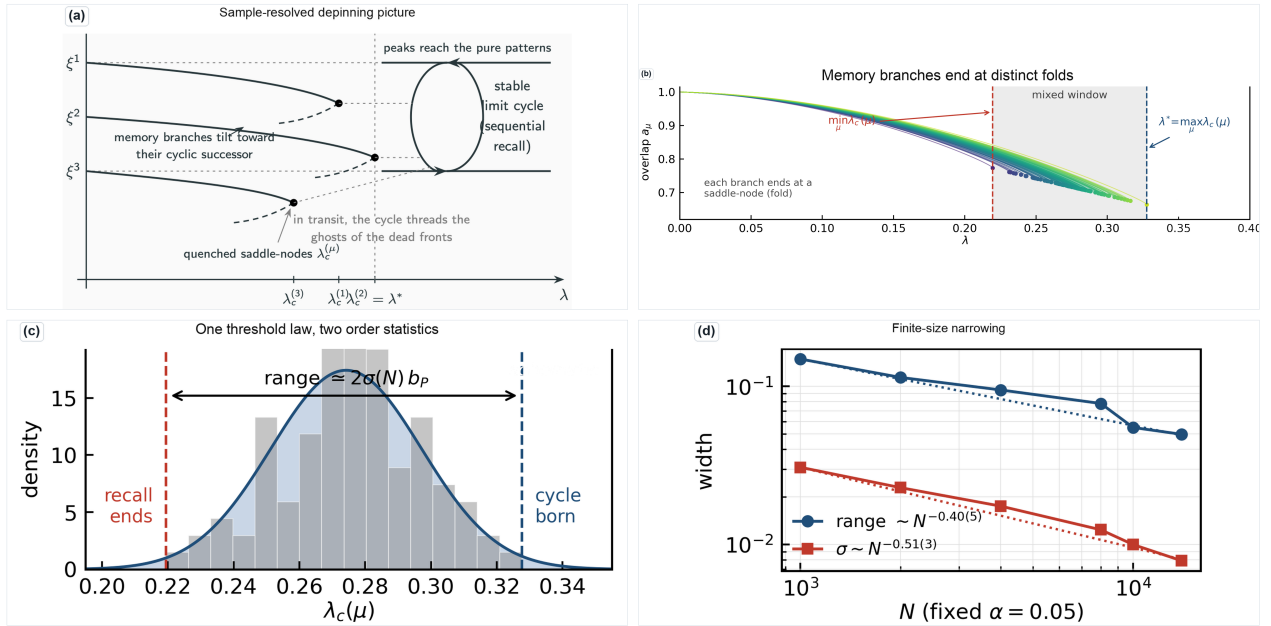


Figure 4: From local folds to a global recall cycle. (a) Schematic organization: fixed-point branches disappear at distinct thresholds $\lambda_c(\mu)$, while a moving orbit follows the route left by their remnants and is released only after the terminal barrier has vanished. (b) Self-overlap along all $P = 100$ primary branches of the reference network; their endpoints fill a finite interval. (c) The same endpoints viewed as a distribution. A typical quantile marks the loss of extensive static recall, whereas the largest threshold predicts the release of global motion. (d) At fixed $\alpha = 0.05$, both the standard deviation and the total range narrow as the system size increases. Panels (b,c): $(N, P, \beta, \tau/t_0) = (2000, 100, 20, 10)$, seed 42.

determinant identity reduces all nontrivial roots to a $P \times P$ determinant,

$$\det \left[(t_0 z + 1)I_P - ((1 - \lambda)I_P + \lambda e^{-z\tau} S) \frac{XDX^T}{N} \right] = 0, \quad (24)$$

while the remaining $N - P$ transverse roots are the harmless modes $z = -1/t_0$.

At $z = 0$, Equation (23) becomes the negative of the ordinary fixed-point Jacobian. A zero characteristic root is therefore exactly the zero real eigenvalue of a fold. I used this identity as a discriminator: continuation locates the fold, and the delayed spectrum checks that no complex pair crossed first. Across the tested ranges of load and delay, the memory-connected branches ended with a real zero mode and no resolved preceding Hopf instability. Near $\alpha \simeq 0.10$, close to the classical retrieval-capacity boundary, the single-sample landscape becomes too rough for precise quantitative thresholds; the low-load regime $\alpha \leq 0.05$, which contains the main results of this report, is cleanly resolved.

4.2 ONE MEMORY, ONE THRESHOLD

Starting from each pattern at $\lambda = 0$, I continued the associated stable retrieval state and its neighboring index-one saddle until they collided. For the reference network, the 100 resulting thresholds span

$$\min_{\mu} \lambda_c(\mu) = 0.2195, \quad \text{median}_{\mu} \lambda_c(\mu) \simeq 0.2742, \quad \max_{\mu} \lambda_c(\mu) = 0.3276. \quad (25)$$

Each endpoint is bracketed to within approximately 2×10^{-4} . The observed range, about 0.11, is therefore a physical spread, not numerical uncertainty. Pattern 28 is the first to disappear and pattern 91 the last. These values coexist inside the same network: the distribution is an internal random object of one quenched sample, rather than sample-to-sample noise around a unique critical point.

The branch deformation and the approach of the leading eigenvalue to zero are shown in Figure 5.

The branch does not remain a pure pattern and then suddenly disappear. As λ grows, the successor interaction progressively twists memory μ toward memory $\mu + 1$. If $\lambda_{50}(\mu)$ denotes the point at which this

Static folds reshape memories before they disappear

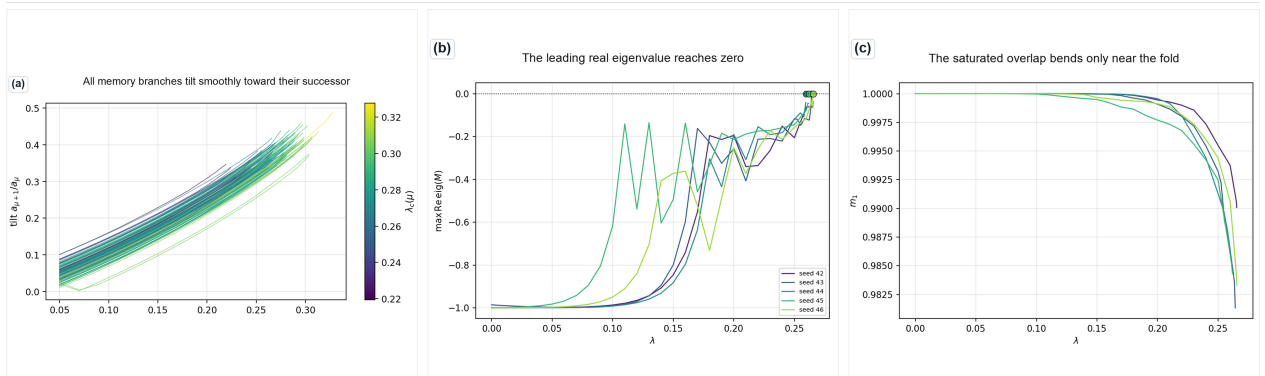


Figure 5: How a memory deforms before its fold. (a) The successor-to-current amplitude ratio rises smoothly on all 100 branches; color denotes the individual threshold. (b) At $N = 10^4$, the leading static eigenvalue reaches zero in five disorder samples. (c) The conventional saturated overlap remains close to one until very near the fold, even though the underlying pattern-space amplitudes have already tilted substantially. Panel (a): reference network; panels (b,c): $\alpha = 0.05$, $\beta = 20$.

successor component has reached half its terminal value, then

$$\text{median}_\mu \frac{\lambda_{50}(\mu)}{\lambda_c(\mu)} = 0.64, \quad \text{IQR} = [0.61, 0.66]. \quad (26)$$

Thus the deformation occupies most of the branch. After rescaling λ by each branch's own threshold, the 100 deformation curves nearly collapse: disorder mainly chooses *when* the branch dies, while its normalized route to the fold is common. The remaining variation is strongly related to the realized overlap between a pattern and its successor.

This deformation can be almost invisible in the ordinary overlap. At high gain, $\tanh(\beta u_i)$ is saturated, so a considerable change in field amplitude may leave every neural sign unchanged. The pattern-space coordinates therefore reveal a broad precursor that the binary memory readout detects only near the endpoint. Close to that endpoint, both the node–saddle separation and the soft real eigenvalue follow the square-root behavior expected from the saddle-node normal form (10).

4.3 THE PROBABILITY LAW OF THE THRESHOLDS

The thresholds are not inserted as independent random numbers; they are derived from the same pattern matrix and live around a cyclic memory chain. Nevertheless, their one-point statistics are remarkably simple. For each sample, define

$$z_\mu = \frac{\lambda_c(\mu) - \bar{\lambda}_c}{\sigma_{\lambda_c}}, \quad (27)$$

where the mean and standard deviation are taken over memories within that sample.

The distributional collapse, size scaling, and extreme-value comparison are collected in Figure 6.

At $N = 10^4$, changing α from 0.01 through 0.02 to 0.05 moves the median threshold from approximately 0.38 to 0.35 and 0.27, while the standardized laws remain statistically compatible. Pooling the well-resolved data gives a maximum CDF distance 0.026 from a fitted normal distribution. Small skewness and excess tails remain, so “near-Gaussian” is the appropriate description, but a Gaussian accurately captures both the bulk and the leading extreme behavior. Neighboring thresholds are also nearly uncorrelated: at $N = 10^4$, their lag-one correlation is -0.009 .

The empirical law can therefore be summarized as

$$\lambda_c(\mu) \approx \bar{\lambda}_c(\alpha) + \sigma_{\lambda_c}(N, \alpha) Z_\mu, \quad (28)$$

where Z_μ has an approximately load-independent standardized distribution. The load sets the center because it controls the mean amount of cross-talk. The width is chiefly a finite-size effect and narrows as

$$\sigma_{\lambda_c}(N) \propto N^{-b}, \quad b = 0.43, \quad 95\% \text{ CI} = [0.40, 0.47] \quad (\alpha = 0.05). \quad (29)$$

This is consistent with a quasi-central-limit mechanism: every branch samples many weak random cross-overlaps, whose combined influence becomes narrower as N increases. The measured exponent is close to, but not assumed to equal exactly, the ideal $1/2$ value.

Thresholds form a finite-size random ensemble

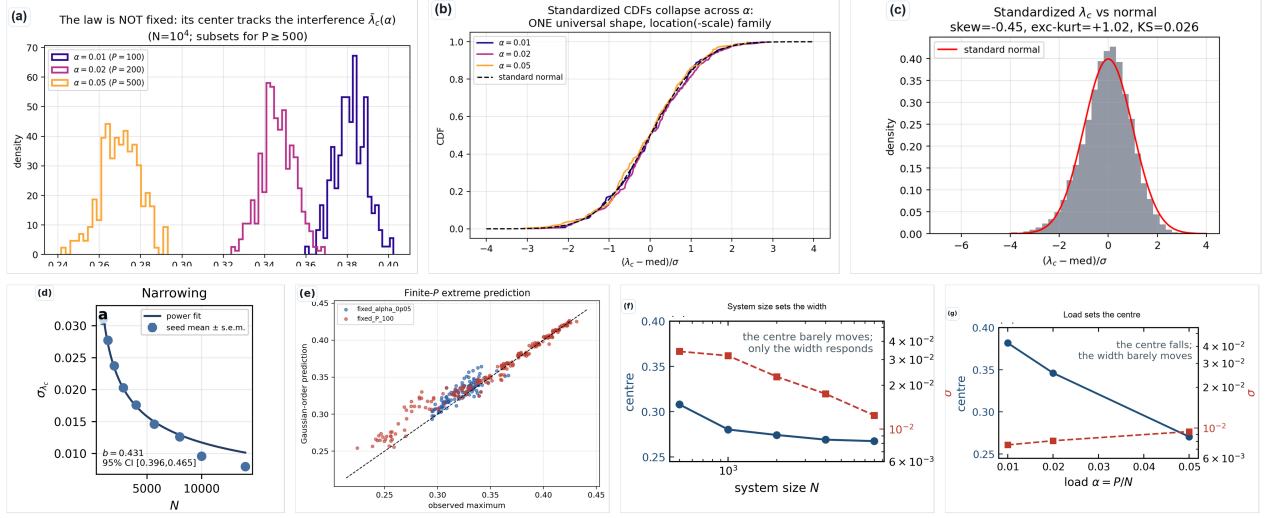


Figure 6: The finite-size law of memory thresholds. (a) At $N = 10^4$, increasing the load shifts the raw distribution. (b,c) After centering and rescaling, empirical distributions at different loads collapse onto a common near-Gaussian shape. (d) A disorder-seed-aware fit gives $\sigma_{\lambda_c} \propto N^{-0.43}$, with 95% confidence interval $[0.40, 0.47]$, consistent with a quasi-central-limit narrowing. (e) The observed sample maxima follow the finite- P Gaussian extreme-value estimate. (f,g) Size mainly controls the width, whereas load mainly controls the center. Uncertainties are resampled over disorder realizations, not over memories of one realization.

4.4 TWO ORDER STATISTICS, TWO PHYSICAL BOUNDARIES

The fraction of original retrieval branches that survive at a given λ is simply the complementary empirical distribution,

$$f_{\text{stat}}(\lambda) = \frac{1}{P} \sum_{\mu=1}^P \Theta[\lambda_c(\mu) - \lambda]. \quad (30)$$

The bulk of the law therefore controls the loss of an extensive number of static memories. A sequence, by contrast, must traverse the entire ring. It remains pinned if even one relevant barrier survives, suggesting that its onset should be governed by $\lambda_* = \max_{\mu} \lambda_c(\mu)$.

For P approximately independent Gaussian draws, standard extreme-value asymptotics give [28]

$$\begin{aligned} \mathbb{E}[\lambda_*] &\simeq \bar{\lambda}_c + \sigma_{\lambda_c} b_P, \\ b_P &= \sqrt{2 \ln P} - \frac{\ln \ln P + \ln(4\pi)}{2\sqrt{2 \ln P}}. \end{aligned} \quad (31)$$

Without an additional fitted parameter, this formula tracks the observed maxima with a mean signed error of 0.00734 and a maximum absolute error of 0.02745, comparable to the natural sample-to-sample fluctuations. This quantitative agreement supports the Gaussian extreme-value interpretation.

At finite N , the typical and maximal thresholds enclose a genuine parameter window. Because $P = \alpha N$ grows linearly while the width narrows approximately as a negative power of N ,

$$\lambda_* - \bar{\lambda}_c \sim \sigma_{\lambda_c} \sqrt{2 \ln(\alpha N)} \longrightarrow 0. \quad (32)$$

Under the measured power-law narrowing, the window therefore closes in the thermodynamic limit. A single macroscopic critical point can conceal two different finite-size events and the internal ensemble that creates them.

Take-home message. Finite quenched disorder assigns each memory its own saddle-node threshold. Those thresholds form a near-Gaussian, quasi-independent ensemble: its bulk controls the loss of extensive static recall, whereas its maximum is the candidate barrier for global sequential motion.

5

FROM THE EXTREME FOLD TO SEQUENTIAL RECALL

The statistical argument of the previous section identifies the largest threshold as a natural candidate for the dynamical onset. It does not prove that a trajectory actually selects this memory, nor that the relevant local bifurcation lies on a closed global route. I therefore compared static continuation and time-dependent simulations as independent measurements. The result is unusually direct: the last static memory is the slowest dynamical relay, and its disappearance releases the sequence.

5.1 THE STATIC MAXIMUM PREDICTS THE DYNAMICAL ONSET

For each disorder realization, I first located λ_* using fixed-point continuation, without simulating a recall cycle. I then followed moving attractors dynamically downward from above onset, asking where motion first stopped and which memory produced the longest dwell time. In the reference network, all three diagnostics select memory 91 and $\lambda_* \simeq 0.328$.

The sample-resolved comparison of these diagnostics is shown in Figure 7.

Across 20 independent networks, the static maximizer, slowest relay, and final arrest memory have the same identity. In one sample the two leading static brackets overlap, so the maximum cannot be uniquely ordered. For each of the remaining 19,

$$\left| \lambda_*^{\text{static}} - \lambda_{\text{onset}}^{\text{dyn}} \right| < 10^{-3}, \quad (33)$$

and the largest retained discrepancy is 7.3×10^{-4} . This agreement is roughly two orders of magnitude finer than the total threshold range. It turns an order-statistical prediction into a sample-resolved dynamical statement.

5.2 A LOCALIZED SADDLE-NODE BOTTLENECK

Near a saddle-node, the normal form predicts complementary square-root laws on the two sides. Below the threshold, the leading real stability eigenvalue of the pinned state must vanish as $(\lambda_* - \lambda)^{1/2}$. Above it, the passage time through the remnant or “ghost” of the destroyed fixed points must grow as $(\lambda - \lambda_*)^{-1/2}$.

Both signatures are observed. Above onset, 99 bonds keep relay times near the sample median while the 91-to-92 passage becomes progressively slower. A representative fit is

$$t_{91 \rightarrow 92} \simeq \frac{C}{\sqrt{\lambda - \lambda_g}}, \quad C = 2.3 \pm 0.2, \quad \lambda_g = 0.3262, \quad (34)$$

The largest static threshold selects collective cycle onset

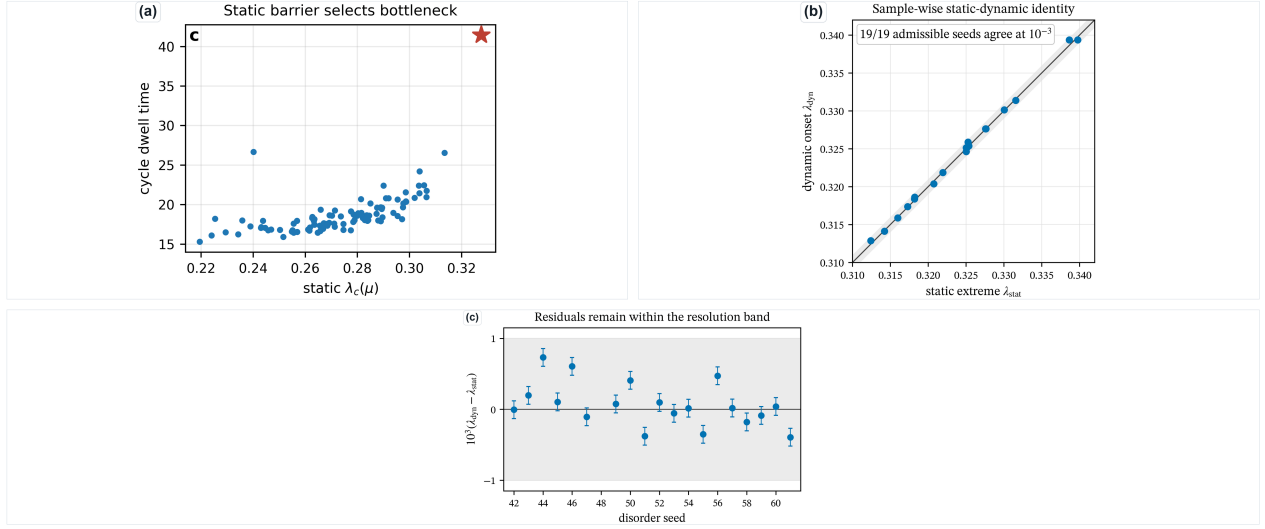


Figure 7: Static extreme selection of the moving onset. (a) In the reference network, the independently measured relay dwell time increases with the continued branch threshold; the largest threshold (star, memory 91) is the bottleneck. (b) Across 19 disorder realizations with a uniquely resolved maximum, the dynamical onset agrees with the static extreme to within the 10^{-3} scan resolution. (c) Seed-resolved residuals and brackets lie within that tolerance. The identity of the terminal memory is correctly selected in all 20 networks tested, with one static tie unresolved inside its continuation bracket. $(N, P, \beta, \tau) = (2000, 100, 20, 10)$.

consistent with the independently bracketed onset $[0.326, 0.328]$. A multi-seed fit on the near-threshold window gives exponent 0.470 with 95% confidence interval $[0.435, 0.504]$, containing the saddle-node value $1/2$ and excluding logarithmic or linear alternatives on the resolved window. Below onset, the leading real eigenvalue softens approximately as

$$\zeta_{\max} \simeq -2.1 \sqrt{\lambda_* - \lambda}, \quad (35)$$

and reaches zero at the same threshold. The last meaningful decade is limited by the uncertainty with which λ_* itself is bracketed, so the diagnosis rests on the resolved scaling window and direct branch continuation, not on arbitrarily close points.

The two-sided slowing, delay collapse, and Floquet evidence are collected in Figure 8.

The moving orbit does not lose transverse stability as onset is approached. Its Floquet spectrum retains the neutral multiplier associated with time translation, while the resolved nontrivial multipliers remain far inside the unit circle. The orbit is lost because its route becomes blocked, not because it becomes weakly attracting. This is the local Floquet signature expected when a saddle-node forms on an invariant circle.

5.3 DIRECT CONSTRUCTION OF THE TERMINAL RECALL ROUTE

Calling the event a SNIC requires a global statement: the colliding node and saddle must belong to a closed invariant connection. I tested this directly in the reference realization. Just below λ_* , I launched trajectories on both sides of the unstable direction of the saddle paired with memory 91. One branch falls onto the neighboring stable node. The other travels forward through the locations of all the already-vanished memories and returns to the same node.

At $\lambda = 0.323$, the tour takes $1991 t_0$ and closes with a pattern-space metric distance 3.5×10^{-14} ; at $\lambda = 0.318$

Critical slowing and terminal depinning

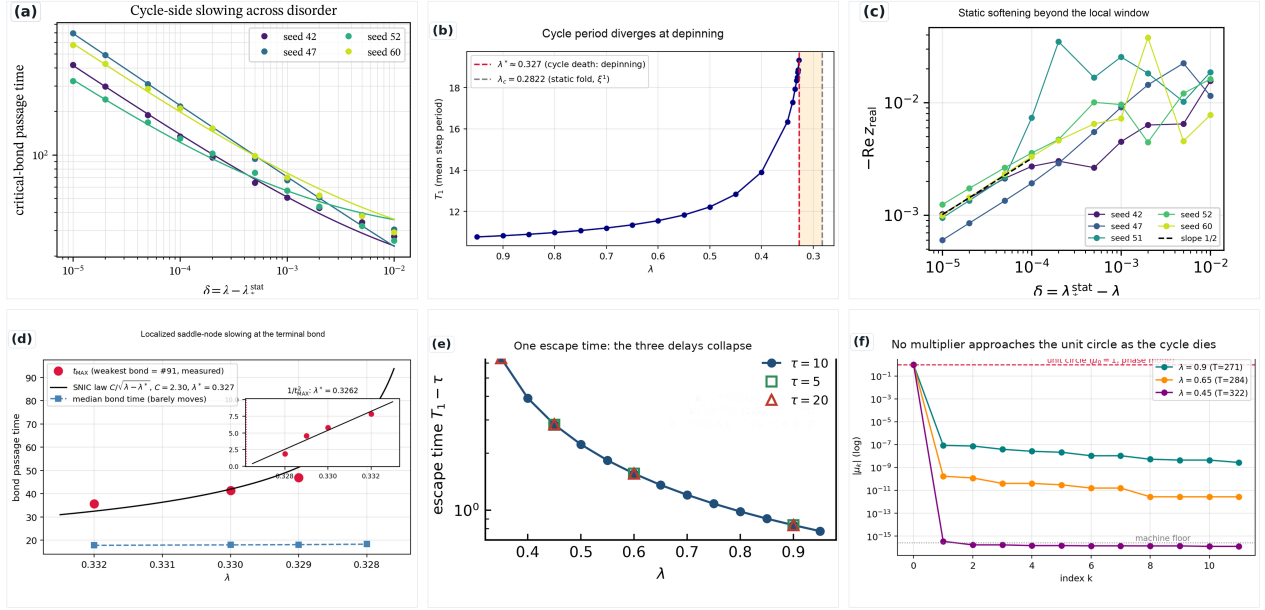


Figure 8: Critical slowing and the terminal depinning event. (a) Cycle-side passage times in four admissible samples show saddle-node-like slowing. (b) A representative static fold and the later depinning threshold delimit the finite-size window. (c) The leading real eigenvalue softens on the static side; points closer than about 10^{-4} to the bracketed threshold are indicative only. (d) The divergence is localized on the terminal bond. (e) Subtracting the explicit delay from the relay period collapses measurements at $\tau = 5, 10, 20$ onto one escape-time curve. (f) Floquet calculations retain the neutral phase direction while all resolved nontrivial directions remain strongly contracting; near numerical floor, plotted values should be read as upper bounds on multiplier magnitude. Together with the directly closed route in Figure 9, these results support a terminal SNIC depinning mechanism. $(N, P, \beta) = (2000, 100, 20)$.

the closure is about 10^{-13} . These residuals are at round-off scale relative to an order-one state. One stage earlier, when memories 43 and 91 remain, the two forward unstable connections link $43 \rightarrow 91$ and $91 \rightarrow 43$. Thus the terminal and penultimate structures are directly closed into the same cyclic route. I did not numerically close the full many-node object at every earlier parameter value, so the strongest justified statement is that the terminal organizing route is an invariant circle, thereby supporting a SNIC at the last fold.

The directly reconstructed route and the orbit released above onset are shown in Figure 9.

5.4 ONE GLOBAL ORBIT, REACHED FROM EVERY MEMORY

Above λ_* , I initialized the dynamics independently at each of the 100 stored patterns. Every trajectory became periodic, visited all memories in cyclic order, and selected memory 91 as its slowest relay. After phase alignment, the waveforms coincide to below one percent of their order-one signal. The quantitative comparison is reported in Table 2. This establishes one global attractor for the tested memory initial conditions. It does not imply that no other attractor exists elsewhere in the full basin structure.

The orbit also recalls the binary patterns more accurately than the tilted static states suggest. At $\lambda = 0.330$, its median analog peak is 0.992 and its smallest is 0.978, while

$$b_\mu(t_\mu^{\text{peak}}) = 1 \quad \text{for every } \mu = 1, \dots, P. \quad (36)$$

The recall orbit persists as a single invariant circle

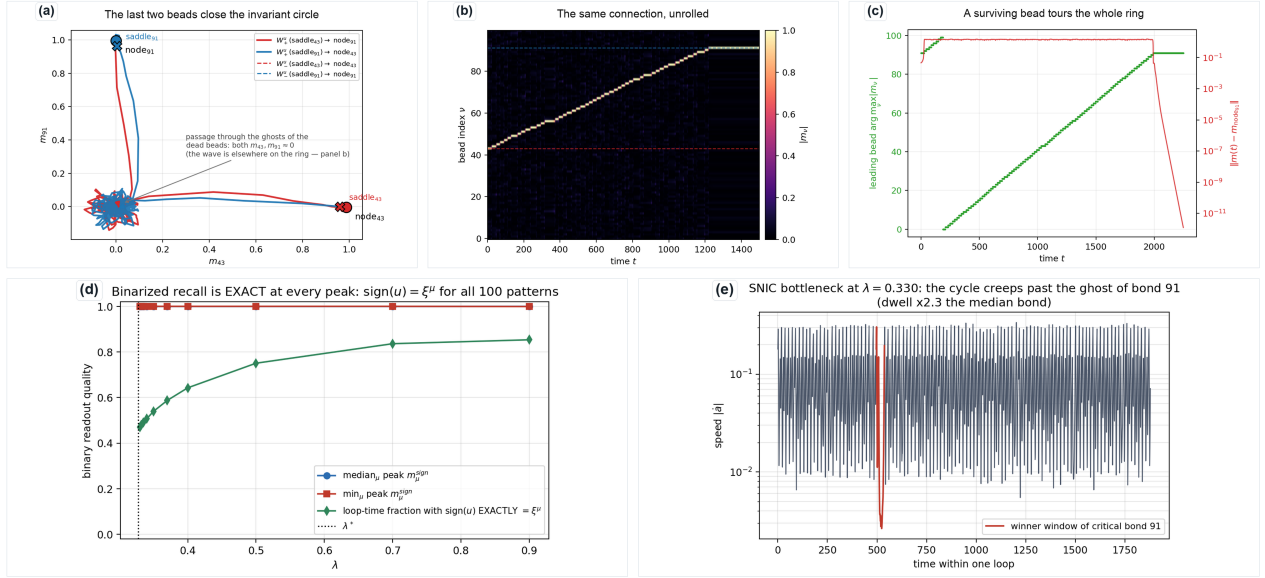


Figure 9: The invariant recall route and the released cycle. (a) When two stable memories remain, the forward unstable branches connect them in both directions around the ring. (b) Unrolling memory index reveals a single wave threading the ghosts of 98 vanished branches. (c) With only memory 91 left, the forward branch tours the full ring and returns to the node at round-off accuracy. (d) Above onset, every memory reaches binary overlap one at its own peak; the green curve gives the fraction of a tour spent in an exact stored pattern. (e) The velocity minimum localizes at the ghost of terminal bond 91. Reference network.

Table 2: Convergence from all stored memories to one recall cycle. $d_{\max}$ is the largest phase-aligned waveform distance from a reference trajectory.

λ	initial memories	mean tour period $\bar{T}/t_0$	$d_{\max}$
0.330	100	1869.4	5.3×10^{-3}
0.350	100	1633.5	9.2×10^{-3}

At every peak, every neuron has the sign of the corresponding stored pattern.

The delay explains this purification. A fixed point sees its own past and is therefore continually pushed toward its successor, which causes the static tilt. On the moving cycle, when memory μ peaks, the delayed state still resembles $\mu - 1$. The successor map sends that delayed state toward μ itself, aligning the instantaneous and delayed fields. Only later does the delayed state become μ , at which point it pushes the network toward $\mu + 1$. Each relay consequently contains a consolidation stage and an escape stage.

This causal decomposition yields the measured clock law

$$T_1(\lambda, \tau) = \tau + t_{\text{esc}}(\lambda). \quad (37)$$

For $\tau = 5, 10, 20$, subtracting τ collapses the remaining escape times to three significant digits. The architecture sets the basic tempo, while the extreme quenched fold makes one relay arbitrarily slow near onset.

Take-home message. The maximum static threshold is not merely correlated with the transition: it selects the same terminal memory and the same onset as the dynamics in every resolved sample. Its saddle-node ghost is the localized bottleneck on a directly closed terminal recall route, and the released orbit performs exact delay-paced

sequence recall.

6

THE FINITE-SIZE WINDOW AND TESTS OF THE MECHANISM

Between a typical fold and the largest fold, most static memories have disappeared but the last barrier still prevents a periodic tour. This interval is not empty: for the sampled initial conditions it is dominated by irregular motion, coexisting with the few retrieval states that survive. Although this behavior is not the main result of the internship, it completes the finite-size phase portrait and suggests controlled tests of the depinning mechanism.

6.1 A STATIONARY HYPERCHAOTIC REGIME

At $\lambda = 0.31$ in the reference sample, only a few percent of the tested initial conditions settle into one of the residual memories. Most produce long moving or irregular trajectories. Basin fractions depend on how initial conditions are sampled, so I use them only to establish coexistence, not as intrinsic phase-space volumes.

The irregular trajectories are deterministically chaotic. In the reference realization, the two leading finite-time Lyapunov exponents are clearly positive; a third estimate is close to zero, as expected for the neutral flow direction. The representative first eight values are

$$(0.035, 0.011, 0.0015, -0.003, -0.008, -0.022, -0.032, -0.042) t_0^{-1}. \quad (38)$$

Two independent expanding directions are sufficient to classify the attractor as hyperchaotic. The corresponding Kaplan–Yorke interpolation is about 6.4, which should be understood as a finite-window effective dimension, not as a thermodynamic or extensive quantity. A separate five-seed calculation recovered multiple positive directions in every realization, demonstrating that the phenomenon is not tied to one pattern draw.

The multi-seed Lyapunov spectrum and the complementary time-series diagnostics are collected in [Figure 10](#).

The time series supplies two complementary checks. Its power spectrum is broadband, without the discrete frequency comb expected from periodic or quasiperiodic motion. Its two-time correlations depend on the lag but, within the resolved waiting-time windows, not on the age of the trajectory. At $\alpha = 0.05$, the spread between normalized correlation curves at lag $20t_0$ is below one percent. The appropriate conclusion is an observed stationary, non-aging chaotic regime on these windows, rather than a proof of an asymptotic Sinai–Ruelle–Bowen measure.

Because the gap between bulk and maximum thresholds closes as N grows, the parameter window hosting this regime also closes. The hyperchaotic behavior is therefore a finite-size consequence of the separated order statistics, not an additional phase expected to retain a finite width at $N \rightarrow \infty$.

Finite-time diagnostics of irregular recall

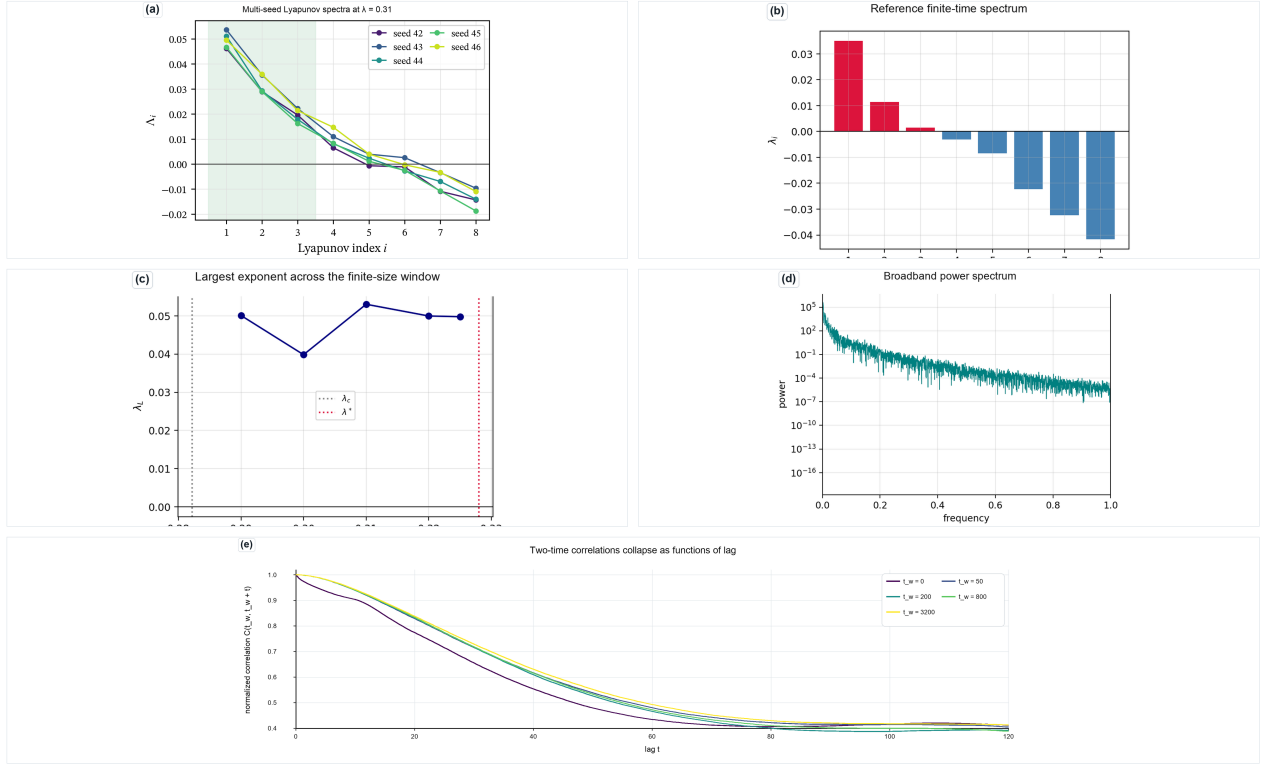


Figure 10: Irregular dynamics in the finite-size bifurcation window. (a) A refined multi-seed tangent calculation finds several positive finite-time exponents at $\lambda = 0.31$. (b) In the reference trajectory, two exponents are strictly positive and the third lies close to the neutral flow direction; the Kaplan–Yorke estimate is $D_{KY} \simeq 6.4$. (c) The largest exponent remains positive through the resolved interval. (d) A representative overlap has a broadband rather than line spectrum. (e) Normalized two-time correlations collapse as functions of lag over the tested waiting times, with no resolved aging. The finite observation windows support a stationary hyperchaotic attractor; they do not constitute a rigorous construction of its invariant measure. $(N, P, \beta, \tau) = (2000, 100, 20, 10)$; panels (b–e) use seed 42.

6.2 TEST I: SHORTENING THE DELAY DELOCALIZES THE FRONT

The extreme-barrier picture assumes that the moving state is localized: at a given instant it should represent essentially one memory, so that one difficult link can stop the entire front. Reducing the delay tests this assumption without moving the fixed points, whose thresholds are delay-independent.

At $\lambda = 0.90$ and $\tau = 10t_0$, the sequence is a narrow succession of nearly pure memories. At zero delay, the motion is instead spread over roughly 15 pattern indices; the largest analog overlap is only 0.16 and the participation number is about 21. A sharp crossover occurs near $\tau_c \simeq 1.3t_0$. Beyond it, the front contracts to approximately one pattern and its relay time becomes reproducible. Fits over five disorder realizations locate the crossover between $1.23t_0$ and $1.34t_0$.

The localization crossover and the delocalized zero-delay trajectory are displayed in Figure 11.

The relay clock changes at the same point:

$$T_1 \simeq \begin{cases} 0.52\tau + 1.73t_0, & \tau \lesssim t_0, \\ \tau + 0.815t_0, & \tau \gtrsim 2t_0. \end{cases} \quad (39)$$

A finite delay synchronizes the propagating recall front

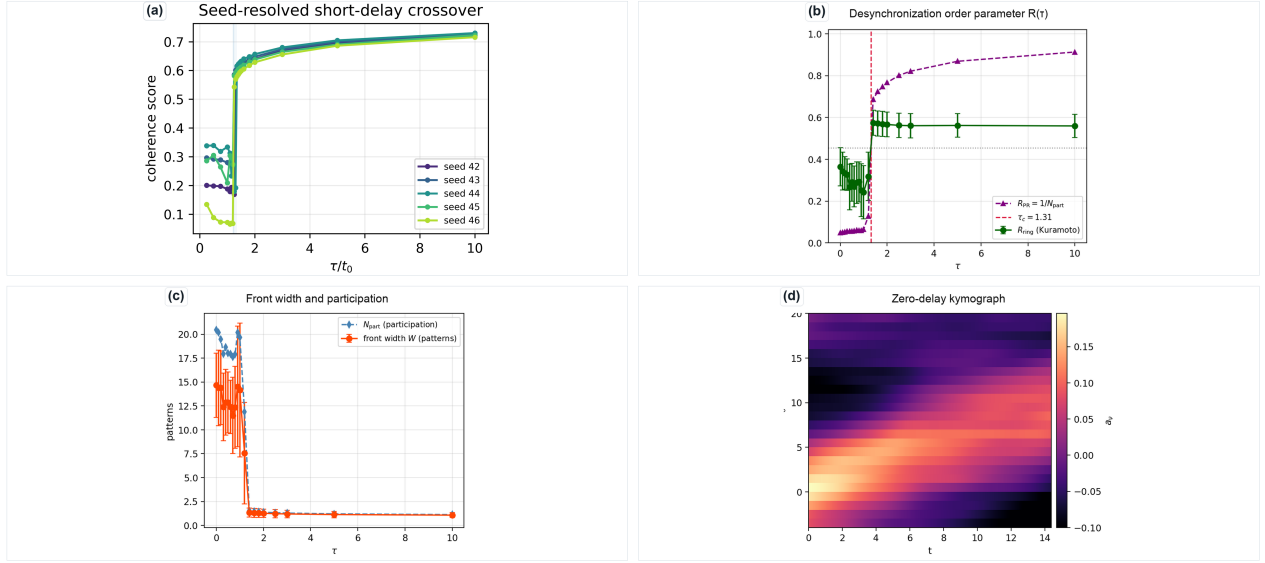


Figure 11: Finite propagation time localizes the recall front. (a) A seed-resolved coherence score rises near $\tau \simeq 1.3t_0$ in five disorder realizations. (b) Independent participation and circular-order diagnostics locate the same crossover. (c) The number of participating memories and the front width collapse from a broad packet to an approximately single-memory wave. (d) At zero delay, the overlap kymograph shows a delocalized packet rather than clean relays. $(N, P, \alpha, \beta, \lambda) = (2000, 100, 0.05, 20, 0.9)$.

At long delay, this recovers the additive law of Equation (37); at short delay, there is no clean one-memory clock. The fixed-point barriers have not moved, but the extended front no longer encounters them one by one. The single-extreme bottleneck then ceases to organize the moving state.

6.3 TEST II: CORRELATIONS RELOCATE OR REMOVE THE BARRIERS

The second test modifies the memory geometry rather than the propagation time. I first generated a correlated sequence by flipping each neural component with probability $(1 - c)/2$ from one frame to the next,

$$\xi_i^{\mu+1} = \xi_i^{\mu} s_i^{\mu}, \quad \Pr(s_i^{\mu} = -1) = \frac{1 - c}{2}. \quad (40)$$

Consecutive patterns then have average overlap c . Closing this finite Markov chain into a ring leaves one special seam: unlike every other neighbor pair, the last and first frames are uncorrelated.

This seam becomes the terminal barrier. At $N = 10^4$, $P = 500$, and $c = 0.2$, the median bulk threshold falls to 0.062 because similar neighbors are easily pulled into one another, while the uncorrelated seam remains at 0.169. Extreme selection still works, but the extreme is now an architectural defect rather than an ordinary statistical fluctuation.

Figure 12 contrasts this seam-dominated ring with the seam-free construction discussed below. In panels (c,d), the internal label K counts retained Fourier modes (denoted K_F here) and is unrelated to the successor matrix K .

To separate the effect of correlation from the effect of the seam, I also generated each neuron's pattern sequence by thresholding a smooth periodic Gaussian Fourier process. This produces a genuinely seam-free ring. As the sequence becomes smoother, the localized primary memories disappear: at nearest-neighbor overlap

Correlated memories: one construction concentrates pinning, the other removes it

(a,b) Markov flip chain closed into a ring: one uncorrelated seam. (c,d) exactly periodic circular Gaussian process: no seam. $N = 10^4$, $P = 500$, $\beta = 20$, seed 42.

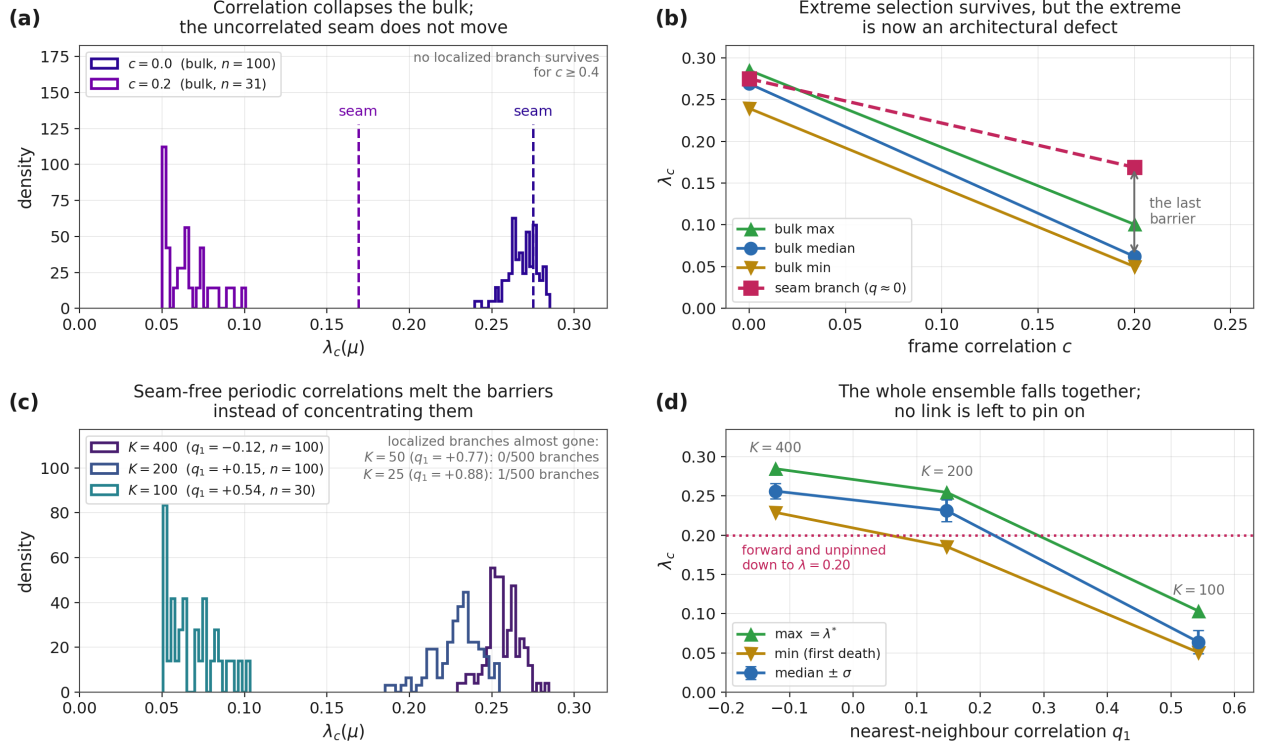


Figure 12: Memory correlations reveal the role of discrete barriers. (a,b) In a Markov sequence closed into a ring, increasing neighbor correlation lowers the bulk thresholds but leaves an uncorrelated seam; the seam becomes the isolated extreme. (c,d) In a seam-free smooth periodic construction, localized primary branches disappear together as the ring becomes smoother. The dynamics remains forward and unpinned down to $\lambda = 0.20$, the lowest value tested. Correlation can therefore concentrate pinning at a defect or remove it altogether, depending on global geometry. $(N, P, \beta) = (10^4, 500, 20)$, seed 42.

about 0.54, only 30 of 500 initialized branches remain identifiable, and at stronger smoothness almost none do. Yet the dynamics remains forward and unpinned down to $\lambda = 0.20$, the lowest value tested. Once the discrete obstacles have melted into a smooth continuous path, there is no exceptional barrier from which a depinning transition could be built.

Together, the two tests isolate the mechanism. A long enough propagation time creates a localized front. Quenched, discrete memory heterogeneity creates distinguishable barriers. Remove the localization and no single barrier controls motion; remove the barriers and there is nothing on which the front can pin.

Take-home message. The finite-size window supports a robust irregular attractor, but the broader lesson comes from controlled perturbations: extreme-value depinning requires both a localized recall front and a discrete heterogeneous set of barriers.

7

REPLACING EXPLICIT DELAY BY NETWORK DEPTH

An explicit point delay makes the mechanism transparent but is an idealization. A strictly Markovian system can nevertheless remember its past through hidden variables: if activity must propagate through several intermediate stages before returning to the monitored layer, an observer of that layer sees an effective delay. I designed a multilayer extension to test whether such architectural depth can replace the delay in the sequence-retrieval mechanism.

7.1 A LAYERED RECURRENT ARCHITECTURE

Let $\mathbf{U}_k \in \mathbb{R}^N$ denote the activity of layer $k = 0, \dots, K_{\text{lay}} - 1$, and define the componentwise response $\phi(\mathbf{U}) = \tanh(\beta \mathbf{U})$. Intermediate layers reconstruct their current state through the Hebbian matrix J and receive activity from the preceding layer,

$$t_0 \dot{\mathbf{U}}_k = -\mathbf{U}_k + J\phi(\mathbf{U}_k) + \lambda_{\text{ff}}\phi(\mathbf{U}_{k-1}), \quad 1 \leq k < K_{\text{lay}}. \quad (41)$$

The last layer closes the loop: its activity is transformed by the successor matrix and fed back into the first layer,

$$t_0 \dot{\mathbf{U}}_0 = -\mathbf{U}_0 + K\phi(\mathbf{U}_{K_{\text{lay}}-1}). \quad (42)$$

The enlarged system is an ordinary differential equation: its future depends only on the current state of all layers. In the numerical analysis below, memory transitions and recall periods are measured on the readout layer $\mathbf{U}_{K_{\text{lay}}-1}$. If the remaining layers are eliminated conceptually, the dynamics of this readout layer alone depends on a distributed history. Depth therefore acts as implicit memory.

The feed-forward propagation path and the recurrent interactions are summarized in Figure 13.

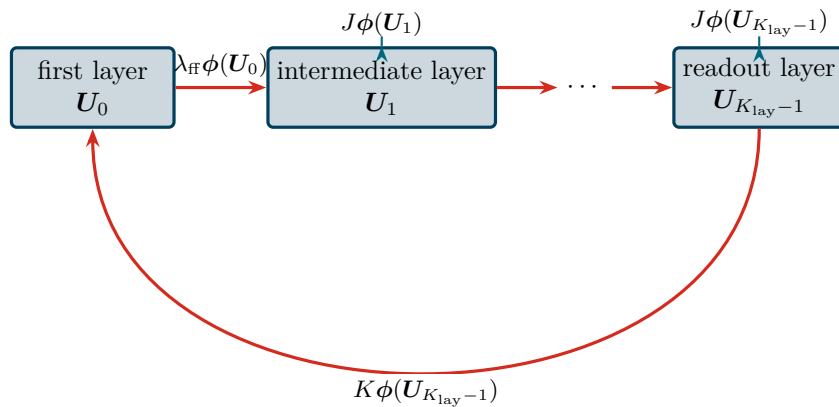


Figure 13: A Markovian architecture with implicit delay. Activity propagates from the first layer through recurrent intermediate layers to the readout layer. The readout activity is then transformed by the successor matrix K and fed back into the first layer, closing the loop. The full multilayer state has no explicit delay, while the readout layer inherits an effective memory from propagation through depth.

7.2 DEPTH CONTROLS THE RECALL PERIOD

I measure the inter-memory recall period from successive memory transitions in the readout layer $U_{K_{\text{lay}}-1}$. For $N = 500$, $P = 25$, and $K_{\text{lay}} = 4, 6, 8, 11$, this period is linear in the number of propagation steps,

$$T_1(K_{\text{lay}}, \lambda_{\text{ff}}) = a(\lambda_{\text{ff}}) + b(\lambda_{\text{ff}})(K_{\text{lay}} - 1), \quad (43)$$

with $R^2 \geq 0.999$ for every fit. Across five disorder realizations,

$$b(1) = 1.67 \pm 0.08, \quad b(2) = 1.17 \pm 0.02, \quad b(4) = 0.94 \pm 0.01, \quad (44)$$

where the units are t_0 per added layer. Two independent propagation-time measurements—the cross-correlation shift between layers and the lag between matched relay events—agree with these slopes to within 3.4%. The clock is therefore carried by physical propagation through hidden variables, not inferred only from the final output period.

The linear depth scaling and its agreement with the independent relay estimator are shown in Figure 14.

On a denser coupling grid, the propagation quantum is well summarized by

$$b(\lambda_{\text{ff}}) \simeq 0.75 + 0.91\lambda_{\text{ff}}^{-1.1}. \quad (45)$$

It approaches roughly $0.75t_0$, not t_0 , at strong coupling. A simple switching argument makes this plausible. If a driven hidden unit relaxes from approximately $-\lambda_{\text{ff}}$ toward $+\lambda_{\text{ff}}$, it crosses zero after $t_0 \ln 2 \simeq 0.69t_0$; both its starting value and its target scale with λ_{ff} , so this time remains finite at large coupling. The measured plateau is slightly higher, consistent with a finite upstream switching time and residual recurrent pull. This argument is explanatory rather than a formal asymptotic derivation; the numerical result is the robust linear depth law and its saturating per-layer slope over the tested sizes and couplings.

On the coarse regime scan, the multilayer model is stationary for $\lambda_{\text{ff}} \leq 0.5$ and supports an ordered wave for $\lambda_{\text{ff}} \geq 1$, with a narrow irregular band between. These results demonstrate the intended principle: a literal delay operator is unnecessary for sequential recall. Hidden Markovian depth can create the propagation time that localizes and clocks the memory wave.

The next step is not merely to repeat the period measurement at larger N . The central open question is whether the same sample-resolved fold ensemble and extreme-barrier selection found in the explicit-delay system

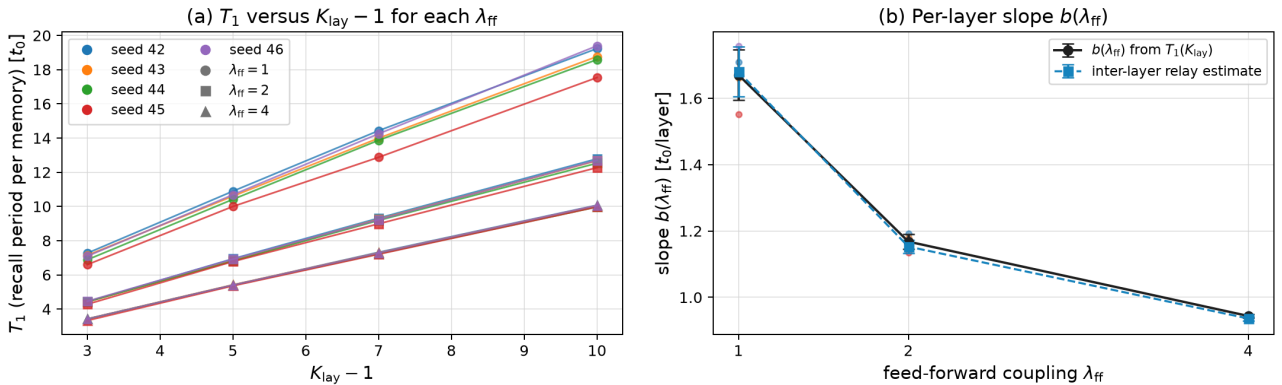


Figure 14: Architectural depth generates the sequence clock. Across five disorder realizations, the recall period grows linearly with $K_{\text{lay}} - 1$. The fitted slope agrees with independently measured inter-layer propagation delays. As the feed-forward coupling λ_{ff} grows, the delay quantum per layer decreases and approaches approximately $0.75t_0$.

persist after the hidden layers have been eliminated. Establishing this would elevate the multilayer construction from a clocking analogy to a full architectural realization of the depinning mechanism.

Take-home message. The multilayer architecture, developed during this internship, replaces a prescribed delay with a measurable propagation time. Its recall period is linear in depth, showing that implicit dynamical memory can generate the sequence clock.

8

CRITICAL ASSESSMENT AND FUTURE DIRECTIONS

The strongest result of this internship is a coherent chain of independently measured facts: each memory has a fold; the folds form an intra-sample distribution; the distribution's maximum selects a specific memory; and static continuation predicts, realization by realization, where the moving orbit is released and where it slows down. This chain makes the finite-size order-statistical mechanism robust. It also helps distinguish what has been established directly from what remains an interpretation or an extrapolation.

8.1 WHAT THE NUMERICAL EVIDENCE ESTABLISHES

Several conclusions are supported by multiple independent diagnostics.

- **Memory-specific folds.** The equilibrium branch, adjacent saddle, real zero mode, and square-root collision identify saddle-node termination. The threshold spread is far larger than the continuation uncertainty and is reproduced over sizes and disorder realizations.
- **Bulk-to-extreme selection.** The static survival fraction reads the bulk of the threshold law, while the cycle onset agrees with the static maximum to 10^{-3} in every realization whose maximum is uniquely resolved. The selected memory identity also matches the slowest relay and final arrest.
- **Near-Gaussian threshold statistics.** Standardized distributions collapse across the tested loads, neighboring thresholds are nearly uncorrelated, and Gaussian finite- P extreme-value theory predicts the observed maxima at the scale of their natural sample fluctuations.
- **A terminal SNIC mechanism in the reference network.** Static node-saddle coalescence, two-sided saddle-node slowing, strong transverse cycle stability, and direct numerical closure of the terminal and penultimate connections jointly support depinning on an invariant circle.
- **Delay-paced exact recall.** The cycle reaches binary overlap one for every stored pattern, and periods at different delays obey the additive law $T_1 = \tau + t_{\text{esc}}(\lambda)$.
- **Implicit delay from depth.** The multilayer system produces ordered replay with a recall period linear in the number of propagation layers, and the slope matches independently measured inter-layer lags.

The finite-size conclusion is especially clear. Under the measured power-law narrowing, the threshold width decreases with N while the number of memories grows only linearly. The bulk and maximum thresholds therefore merge as expressed in Equation (32); the separated window is a finite-size phenomenon whose mechanism becomes hidden in the thermodynamic description.

8.2 LIMITATIONS OF THE PRESENT STUDY

The project was designed as a numerical-physics investigation, and its main limitations follow from that choice.

First, the threshold distribution does not yet have a first-principles analytical derivation. The quasi-central-limit explanation is physically persuasive—each branch samples many weak cross-overlaps—and is supported by the near-Gaussian collapse, weak lag correlations, and size narrowing. It does not yet predict the load-dependent center, the finite-size variance, or the observed deviations from an exact Gaussian.

The detailed threshold statistics were established in a controlled low-load regime, principally for $\alpha \leq 0.05$ at $\beta = 20$. As the load approaches the classical storage-capacity region, the fixed-point landscape becomes increasingly rough and sample-to-sample fluctuations grow, making a branch-resolved description less reliable. The present results therefore do not establish whether the near-Gaussian threshold law, its finite-size narrowing, and the same bifurcation mechanism persist up to capacity. This is precisely the regime in which a disorder-averaged dynamical mean-field treatment would become most valuable.

Second, the global invariant circle was closed directly at the terminal and penultimate stages of the reference realization. The seed-resolved agreement of static and dynamic thresholds and the multi-seed slowing laws strongly support the same local depinning scenario more broadly, but a full global connection was not constructed in every sample or at every earlier fold. This is why I describe the evidence as a terminal SNIC in the reference network and as a consistent depinning mechanism across the ensemble.

Third, all claims about chaos are necessarily finite-time numerical statements. Hyperchaos is supported by multiple positive Lyapunov directions and a broadband spectrum, and the correlation functions show no aging over the sampled waiting times. Longer integrations or more formal invariant-measure techniques would be needed to turn the observed stationarity into an asymptotic theorem. This limitation does not affect the main bulk-to-extreme mechanism, because hyperchaos is a consequence of the finite-size window rather than the evidence used to locate its boundaries.

Finally, the layered architecture establishes an implicit recall clock, not yet the full equivalence of mechanisms. I have not continued all its memory branches or shown that its ordered onset is selected by the maximum of an analogous fold distribution. That is the most direct unfinished question created by this internship.

8.3 EXTENDING THE IMPLICIT-DELAY MODEL

The layered architecture suggests a concrete next numerical campaign: continue every memory-connected equilibrium as the feed-forward coupling λ_{ff} is varied, measure its threshold distribution, and compare its maximum with the ordered-wave onset. If the same selection law holds, depth will have reproduced not only the explicit delay's clock but also its quenched depinning mechanism. If it does not, the failure will reveal which feature of a point delay (rather than memory in general) is essential.

More broadly, the same analysis could be applied to sparse or spatially local networks, to several stored cycles sharing memories, and to controlled levels of dynamical noise. The transferable diagnostic is simple: resolve local bifurcations inside one disordered sample, measure their distribution and correlations, and determine which order statistic each collective observable follows.

Take-home message. The numerical evidence securely supports sample-resolved extreme-barrier selection. The main open tasks are analytical: derive the threshold law and determine whether the multilayer implicit-delay architecture reproduces the same bifurcation ensemble rather than only the same clock.

9

CONCLUSION

This internship began with a classical problem in associative memory: how a recurrent network can preserve static patterns while also replaying them as a sequence. Symmetric Hebbian couplings, asymmetric successor couplings, and a slow response were already known to provide the required ingredients. My contribution was to examine the transition they produce inside individual finite disordered networks, memory by memory and bifurcation by bifurcation.

The resulting picture is more structured than a conventional phase boundary suggests. Every stored memory is connected to a fixed-point branch that twists toward its successor and ends at a memory-specific saddle-node threshold. Within one realization, the thresholds form a near-Gaussian and almost uncorrelated ensemble. The load primarily shifts its center, while increasing system size narrows its width in a quasi-central-limit fashion. Different collective questions then read different order statistics of this one law. A typical quantile marks the loss of an extensive population of static memories; the maximum identifies the last barrier that must disappear before a trajectory can tour the full sequence.

The connection between statistics and dynamics is sample-resolved. Across the tested disorder realizations, static continuation predicts the identity and parameter value of the terminal dynamical bottleneck. Near that bottleneck, a single relay exhibits inverse-square-root slowing, the pinned state develops a soft real mode, and the periodic orbit remains transversely attracting. In the reference realization, the terminal and penultimate unstable connections close directly into an invariant recall route. Once the last node–saddle pair disappears, this route becomes a stable global cycle reached from the initial conditions corresponding to all 100 memories. The delay does more than slow the motion: it realigns the instantaneous and delayed fields, allowing every binary pattern to be recalled exactly and setting the relay clock through $T_1 = \tau + t_{\text{esc}}$.

The interval between the bulk and the extreme is a genuinely finite-size dynamical window, populated by a stationary hyperchaotic attractor alongside the few remaining retrieval states. Controlled modifications clarify the mechanism. A short delay spreads the front over many memories, so one bond no longer controls its arrest. Smooth periodic correlations remove the discrete barriers, while a correlation seam concentrates the extreme at a single architectural defect. The mechanism therefore requires both a localized traveling state and quenched heterogeneous obstacles.

Finally, the multilayer extension developed shows that the propagation time need not be imposed as an explicit delay. A larger Markovian network can store history in hidden layers, and its sequential-recall period grows linearly with depth. Establishing whether this implicit-memory architecture also reproduces the same threshold ensemble is a natural continuation of the project.

As N grows, the threshold distribution narrows and the typical and extreme boundaries merge. The thermodynamic limit retains one critical point but hides how it was assembled. The broader lesson is therefore methodological: in a finite disordered system, local states that are equivalent after averaging may carry an internal ensemble of bifurcations. Collective observables that depend on a typical local event or on the completion of all local events will select different parts of that ensemble. Resolving those quenched bifurcations turns finite size from a nuisance into the regime where the mechanism of the phase transition can actually be seen.

I intend to develop the research presented in this report into a scientific article under Prof. Vitelli's supervision.

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